

1. python NaiveBayes.py

```
/Users/nicholasmccracken/Repos/College/IntroAI_3521/HW03_Programming_3521/NaiveBayes/NaiveBayes.py:148: RuntimeWarning: divide by zero encountered in log
```

```
log_prob += np.log(self.conditional[label][idx])
```

Total Sentences correctly: 775

Predicted correctly: 565

Accuracy: 72.90323%

2. python NaiveBayes_improved.py

Total Sentences correctly: 775

Predicted correctly: 628

Accuracy: 81.03226%

3. Improvement to NaiveBayes

I improved upon the original NaiveBayes prediction method by avoiding the “RuntimeWarning: divide by zero encountered in log” error. To accomplish this, I used a very small value near zero (epsilon variable in NaiveBayes_improved.py code) as a substitute for the log function input whenever the conditional probability of the word in the given class was zero. This increased the accuracy of the algorithm by nearly 10%.